MoleculeLens: Interpretable Drug–Text Retrieval via Salient Molecular Substructures

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Abstract

Drug–text retrieval systems can connect molecular structures to mechanism descriptions, but most current models are difficult to audit. They often rely on large black-box encoders, and benchmark scores can be inflated by memorized scaffolds, drug names, or target metadata. As a result, it is unclear whether a retrieved match is supported by meaningful molecular substructures or by easier lexical shortcuts. We present *MoleculeLens*, an interpretable drug–text retrieval framework that makes each alignment decision traceable to salient molecular features. MoleculeLens aligns fixed molecular fingerprints and frozen biomedical text embeddings using learned linear projection heads. This simple design is the key contribution: the molecule-side linear map gives an efficient saliency score for each ECFP4 bit, allowing retrieval decisions to be inspected at the level of chemically meaningful substructures. On held-out scaffold pairs, this saliency proxy closely matches exact score-derived quantities, with mean Pearson correlation 0.905 against normalized-score gradients and 0.947 against leave-one-bit-out score drops. We evaluate MoleculeLens on a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs using a Bemis–Murcko scaffold-disjoint split. MoleculeLens achieves the strongest Recall@1 and MRR among the evaluated cross-modal baselines on this split, while also exposing when performance depends on drug-name or target/action metadata. Beyond retrieval accuracy, MoleculeLens provides pharmacophore-level explanations for correct matches, faithful diagnostics for failed retrievals, and a same-model substructure-level analysis of lexical leakage. These results show that lightweight linear bridges can make drug–text alignment more auditable without giving up competitive top-rank retrieval performance.

1 Introduction

Drug–text alignment asks whether molecular structures and mechanism descriptions can be embedded in a common space for retrieval. The problem is useful for mechanism search, dataset curation, and model auditing, but it is hard to trust a high retrieval score by itself. Drug names, target labels, and scaffold-correlated train/test examples can make a model look accurate even when the molecular evidence behind a match is weak. Standard dual encoders also give little direct evidence about which substructures support a retrieved mechanism.

MoleculeLens addresses this gap with a deliberately constrained design. It freezes the molecule and text encoders, learns only thin linear projection heads, and uses the molecule-side linear map as a lens on ECFP4 fingerprint bits. The goal is not to build the largest drug–language model. The goal is to make drug–text retrieval auditable: each prediction can be traced back to chemically meaningful fingerprint environments, and shortcut behavior can be inspected at the substructure level.

Contributions.

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- **An auditable retrieval architecture.** MoleculeLens aligns fixed molecular fingerprints and frozen biomedical text embeddings with learned linear bridges, so retrieval remains competitive while the molecular side stays directly inspectable.
- **Validated substructure saliency.** The linear molecular bridge yields a closed-form ECFP4 saliency proxy that closely matches exact normalized-score gradients and exact leave-one-bit-out score drops on held-out scaffold pairs.
- **A scaffold-disjoint drug–mechanism benchmark.** We construct a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs and evaluate under Bemis–Murcko scaffold shift, with explicit controls for drug-name and target/action metadata.
- **Audits beyond top-line accuracy.** We use the same lens to explain correct matches, diagnose failed retrievals, quantify same-model drug-name leakage as substructure shifts, and localize drug-retrievable text signal with a descriptive logit-lens diagnostic.

Related work and positioning. Drug–text and molecule–language models include retrieval-oriented systems such as Text2Mol [Edwards et al., 2021], molecular language models such as MolT5 [Edwards et al., 2022] and BioT5 [Pei et al., 2023], graph-language systems such as MolCA [Liu et al., 2023], and multimodal biomedical pretraining such as KV-PLM [Zeng et al., 2022]. Graphormer [Ying et al., 2021] and Uni-Mol [Zhou et al., 2023] are important molecular representation baselines, though not direct text-query retrieval systems without additional alignment. MoleculeLens is complementary to these higher-capacity models: it emphasizes scaffold-disjoint drug–mechanism retrieval and auditable substructure-level explanations. The evaluation design follows the molecular-learning practice of scaffold splits [Bemis and Murcko, 1996] and combines it with text leakage controls. The interpretability component is related to linear feature analysis and attribution-graph work [Bricken et al., 2023, Templeton et al., 2024, Lindsey et al., 2025], but our features are chemically grounded from the start because ECFP4 bits correspond to circular molecular environments [Rogers and Hahn, 2010] recoverable with RDKit [RDKit contributors, 2023].

2 Main Contributions

Strategy. Figure 1 summarizes the central strategy. MoleculeLens does not attempt to interpret every hidden unit of a large multimodal model. Instead, it inserts a small linear bridge between a fixed molecular representation and a frozen biomedical text encoder. That bridge creates a shared retrieval space and, because the molecule-side map is linear, a direct route from each retrieval score back to ECFP4 substructures.

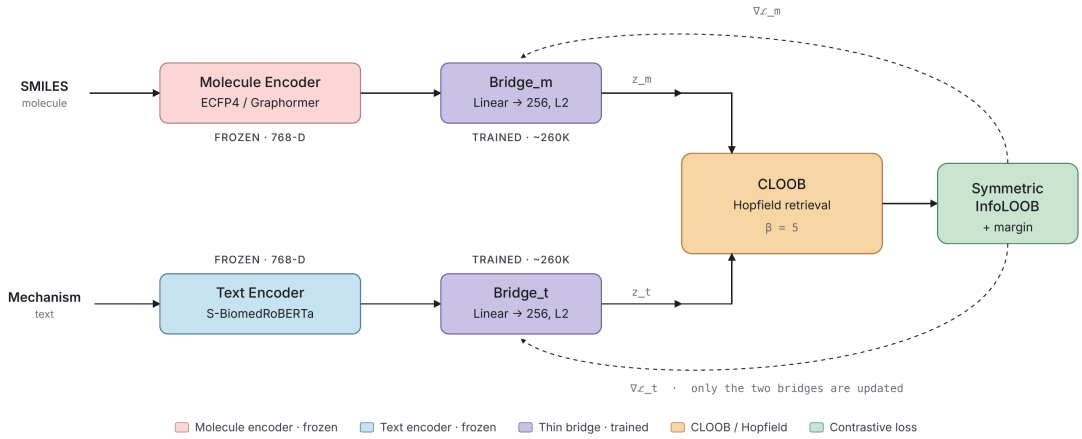

Figure 1: **MoleculeLens strategy.** Frozen ECFP4 and biomedical text encoders are connected by learned linear projection heads and a CLOOB/InfoLOOB retrieval objective. The molecule-side linear map enables direct ECFP4-bit saliency; the text-side projection is reused for the layer-wise logit-lens diagnostic.

Linear bridge retrieval model. For drug i, RDKit computes a radius-2, 2048-bit ECFP4 fingerprint x_i [Rogers and Hahn, 2010]. For text i, a frozen biomedical sentence encoder produces a CLS-style embedding z_i. MoleculeLens learns only two linear projection heads,

$$b_i^{(m)} = W_m x_i + b_m, \qquad b_i^{(t)} = W_t z_i + b_t, \tag{1}$$

followed by L_2 normalization and cosine scoring. Training uses symmetric InfoNCE with same-target weighting and a hardest-same-target margin term. The scaffold model uses batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, margin 0.1, and 100 epochs. All encoders are frozen; only W_m, b_m, W_t, b_t are trained. Training the projection heads takes under 30 seconds per run on a single NVIDIA GPU; ECFP4 attribution and contrastive logit-lens analyses each complete in under 10 minutes. The dominant cost is the one-time frozen text encoding shared across experiments. Additional implementation details are in Appendix B.

Substructure saliency from the molecular bridge. The normalized retrieval score is

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)} = \mathrm{normalize}(W_m x_i + b_m) \cdot \mathrm{normalize}(W_t z_j + b_t). \tag{2}$$

Because W_m is linear, we use the closed-form ECFP4 saliency proxy

$$\mathrm{attr}_k(i,j) = \left[W_m^\top \hat{b}_j^{(t)}\right]_k, \tag{3}$$

where $\hat{b}_j^{(t)} = \mathrm{normalize}(W_t z_j + b_t)$. Equation 3 is a fast ranking proxy rather than an exact causal decomposition: the exact normalized score also depends on the molecule-side bias and $\|W_m x_i + b_m\|$. This is why the experimental section validates it against exact gradients and exact leave-one-bit-out score drops before using it for chemical interpretation.

Diagnostics enabled by the same bridge. The same linear bottleneck supports three audits. First, saliency over present ECFP4 bits lets us inspect correct and incorrect retrievals as substructure-level decisions. Second, removing only the explicit drug-name span at inference while holding model weights fixed tests whether lexical leakage changes which molecular features are salient. Third, applying the trained text bridge to each hidden layer of the frozen text encoder gives a descriptive contrastive logit lens for locating where drug-retrievable text information becomes available.

3 Experimental Validation

Dataset and task. We construct the benchmark from ChEMBL [Zdrazil et al., 2024] by filtering for approved drugs (`max_phase=4`), valid mechanism text, and parseable canonical SMILES. The final dataset contains 2,699 drug–text pairs spanning 1,196 unique Bemis–Murcko scaffolds. The primary text field concatenates mechanism of action, target name, action type, and drug name:

$$\texttt{text_rich} = \texttt{mechanism} + \texttt{target} + \texttt{action} + \texttt{drug}. \tag{4}$$

The main split uses seed 0 over unique scaffolds: 2,019 train examples, 245 validation examples, and 435 held-out test examples whose scaffolds are unseen during training. We report text-query retrieval against a molecule gallery using Recall@1, MRR, Recall@5, and Recall@10 for apples-to-apples comparison with baselines. All main-table rows use single-label diagonal evaluation: a prediction is counted correct only when the exact matching row is returned. This is intentionally conservative for drugs with multiple ChEMBL mechanism rows, but it is the only consistent apples-to-apples protocol for baselines where per-example ranked lists are not available.

Several test drugs appear under multiple target–action rows. For attribution analyses, same-drug and wrong-but-close filtering use RDKit `FragmentParent` standardisation over SMILES to avoid treating salt or formulation variants as independent structures. In the canonical scaffold test set, 193 of 435 rows belong to 76 parent structures with multiple mechanism rows.

3.1 Retrieval and Leakage Controls

Table 1 shows the main comparison. MoleculeLens has the strongest scaffold-split Recall@1 and MRR among the evaluated cross-modal baselines, improving Recall@1 over MolPrompt by 9.6 points and over KV-PLM by 17.0 points. MolPrompt has higher Recall@5/10, so the claim is not leaderboard dominance: MoleculeLens is strongest at top-rank retrieval while remaining competitive at broader candidate coverage. Full-gallery rows are included only as context; the Thin Bridges global result is a memorization-heavy reference, not the main scientific comparison.

Table 1: Primary retrieval comparison. All rows use single-label diagonal evaluation. Full-gallery rows are context only; scaffold rows are the main comparison.

Model	Eval set	R@1	MRR	R@5	R@10
MoleculeSTM (zero-shot)	Full (N=2699)	0.089	0.171	0.243	0.337
Thin Bridges (Global)	Full (N=2699)	0.747	0.849	0.985	0.997
MolPrompt (Global)	Full (N=2699)	0.123	0.264	0.405	0.590
KV-PLM (Global)	Full (N=2699)	0.015	0.038	0.044	0.074
Graphormer (zero-shot)	Scaffold (N=435)	0.002	0.015	0.011	0.025
MoleculeLens	Scaffold (N=435)	**0.225**	**0.304**	0.384	0.462
MolPrompt	Scaffold (N=435)	0.129	0.256	**0.400**	**0.531**
KV-PLM	Scaffold (N=435)	0.055	0.121	0.172	0.251

Text-field and leakage controls. Because `text_rich` includes target/action metadata and an explicit drug field, we report three complementary controls in Table 2. Removing only the drug field and retraining gives R@1 = 0.110, a 51.0% relative drop. Holding the rich model fixed and removing only `Drug: X.` at inference gives R@1 = 0.067, a stricter 70.4% same-model causal diagnostic. Training on mechanism text alone, removing target name, action type, and explicit drug name, gives R@1 = 0.124 and MRR = 0.226. Thus target/action fields contribute real signal, but mechanism-only retrieval remains nontrivial and close to MolPrompt scaffold R@1.

Table 2: Key controls on the canonical scaffold split. Confidence intervals are 95% intervals for Recall@1; MoleculeLens uses exact bootstrap resampling and other baselines use Wilson intervals.

Condition	R@1	MRR	R@1 95% CI
MoleculeLens `text_rich`	0.225	0.304	[0.186, 0.267]
MoleculeLens retrained no-drug	0.110	–	[0.083, 0.140]
MoleculeLens same-model no-name	0.067	–	[0.044, 0.092]
MoleculeLens mechanism-only	0.124	0.226	–
MolPrompt scaffold	0.129	0.256	[0.101, 0.164]
KV-PLM scaffold	0.055	0.121	[0.037, 0.081]

The non-overlapping Recall@1 intervals for MoleculeLens `text_rich` and MolPrompt support the reliability of the top-rank improvement despite the 435-example test set. We also reran the full MoleculeLens objective across the same five consecutive scaffold seeds used by the proxy robustness analysis. Table 3 shows that the complete objective is not weaker than the simplified proxy: mean R@1 is 0.277 ± 0.081 and mean MRR is 0.354 ± 0.083. The higher variance is expected because each seed induces a different scaffold-held-out test set, but every full-loss seed remains above KV-PLM scaffold R@1 and four of five exceed MolPrompt scaffold R@1. The three leakage numbers answer different questions: the 51.0% retrained drop is the fairest model-comparison number, the 70.4% same-model drop is the causal diagnostic for the fixed rich model, and the 22.9% multi-seed proxy mean measures partition sensitivity rather than canonical leakage.

Table 3: Five-seed full-loss check. These runs use the same same-target weighting and hardest-same-target margin term as the main MoleculeLens model. Seeds 0–4 were selected as the same consecutive seed set used by the proxy analysis before evaluating full-loss results.

Seed	n_{test}	R@1	R@5	R@10	MRR
0	403	0.221	0.335	0.424	0.291
1	261	0.295	0.479	0.533	0.381
2	224	0.259	0.375	0.442	0.328
3	221	0.407	0.561	0.629	0.484
4	395	0.203	0.342	0.451	0.284
Mean	–	**0.277**	**0.418**	**0.496**	**0.354**
Std	–	0.081	0.098	0.085	0.083

3.2 Attribution Faithfulness and Substructure Audits

We validate Equation 3 directly against exact gradients and exact leave-one-present-bit-out score drops over all 435 scaffold test pairs before using it as an interpretability tool.

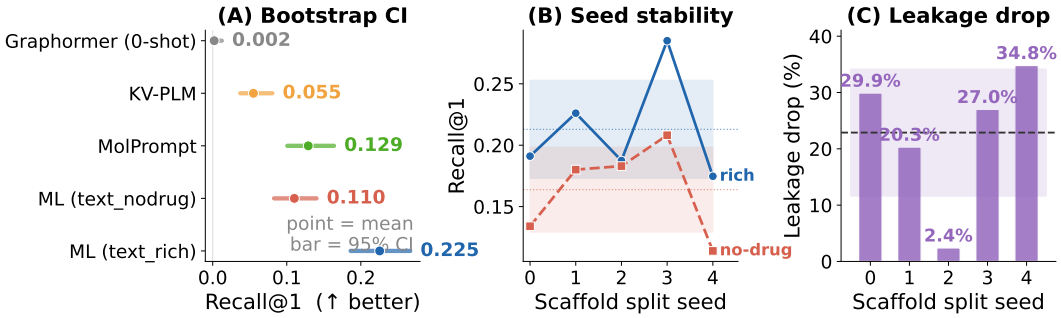

Figure 2: **Statistical robustness summary.** Bootstrap intervals, multi-seed scaffold variance, and per-seed drug-name leakage drops show that the main Recall@1 gap is statistically reliable while leakage varies with scaffold partition.

Table 4: Validation of Equation 3. All statistics are over present ECFP4 bits. The final rows test drug-specificity using distinct-drug pairs with identical no-drug text.

Metric	Value
Mean Pearson(attr, exact gradient)	0.905
Mean Pearson(attr, exact bit-drop)	0.947
Mean top-10 Jaccard(attr, exact bit-drop)	0.710
Incorrect Rank-1 pairs: Pearson(attr, bit-drop)	0.970
Incorrect Rank-1 pairs: top-10 Jaccard	0.761
Identical no-drug text: mean exact top-10 Jaccard	0.136
Identical no-drug text: non-identical exact top-10 sets	952/954

Table 4 supports rank-based use of the saliency proxy. It tracks the exact normalized-score gradient strongly and the exact bit-drop score even more closely. The identical-text test addresses a key conceptual concern: the exact top-10 bit-drop sets are almost never identical across distinct drugs sharing the same no-drug query, so the implemented normalized score is strongly drug-specific. The proxy also remains faithful for failed retrievals, which means incorrect predictions remain auditable decisions of the learned scoring function rather than cases where the attribution method collapses. Appendix D provides additional case studies.

Across these examples and the additional cases in Appendix D, the highest-saliency fingerprint bits align with known pharmacophoric or mechanism-defining substructures, including steroid cores, sulfonamide groups, nucleoside analogue motifs, beta-lactam rings, and iodinated benzoate rings.

What substructures recur? The saliency lens is useful only if it identifies interpretable molecular features rather than a single global shortcut. Appendix Table 7 summarizes the most frequent high-saliency ECFP4 bits across the scaffold test set. Bit 1538 is the most common, appearing in the top-10 for 85 pairs; RDKit decoding maps it to a benzene-ring environment with at least one heteroatom substituent at radius 2, a common pharmacophore anchor across several mechanism families. Frequencies drop sharply after the leading bits, and several frequent bits have near-zero mean signed attribution because they appear in both positive and negative contexts. This pattern supports the intended interpretation: MoleculeLens exposes recurring chemically meaningful substructures, but retrieval decisions are not dominated by one universal fingerprint bit.

Auditing near misses. Faithful attribution is useful beyond correctly retrieved examples. Among rank-2/3 errors after parent-structure filtering, 30 of 43 have attribution correlation above 0.5 between the correct drug and the retrieved drug, with mean correlation 0.569. These are not counted as correct in the diagonal retrieval metric, but the analysis distinguishes chemically plausible near misses from arbitrary failures. Figure 4 shows a representative case: sodium phenylacetate is ranked second while sodium phenylbutyrate is ranked first, and the two molecules share the same ammonia-lowering mechanism with highly overlapping salient substructures. This does not make the retrieval correct, but it shows that MoleculeLens can expose when an error is driven by related chemical evidence rather than an opaque embedding artifact.

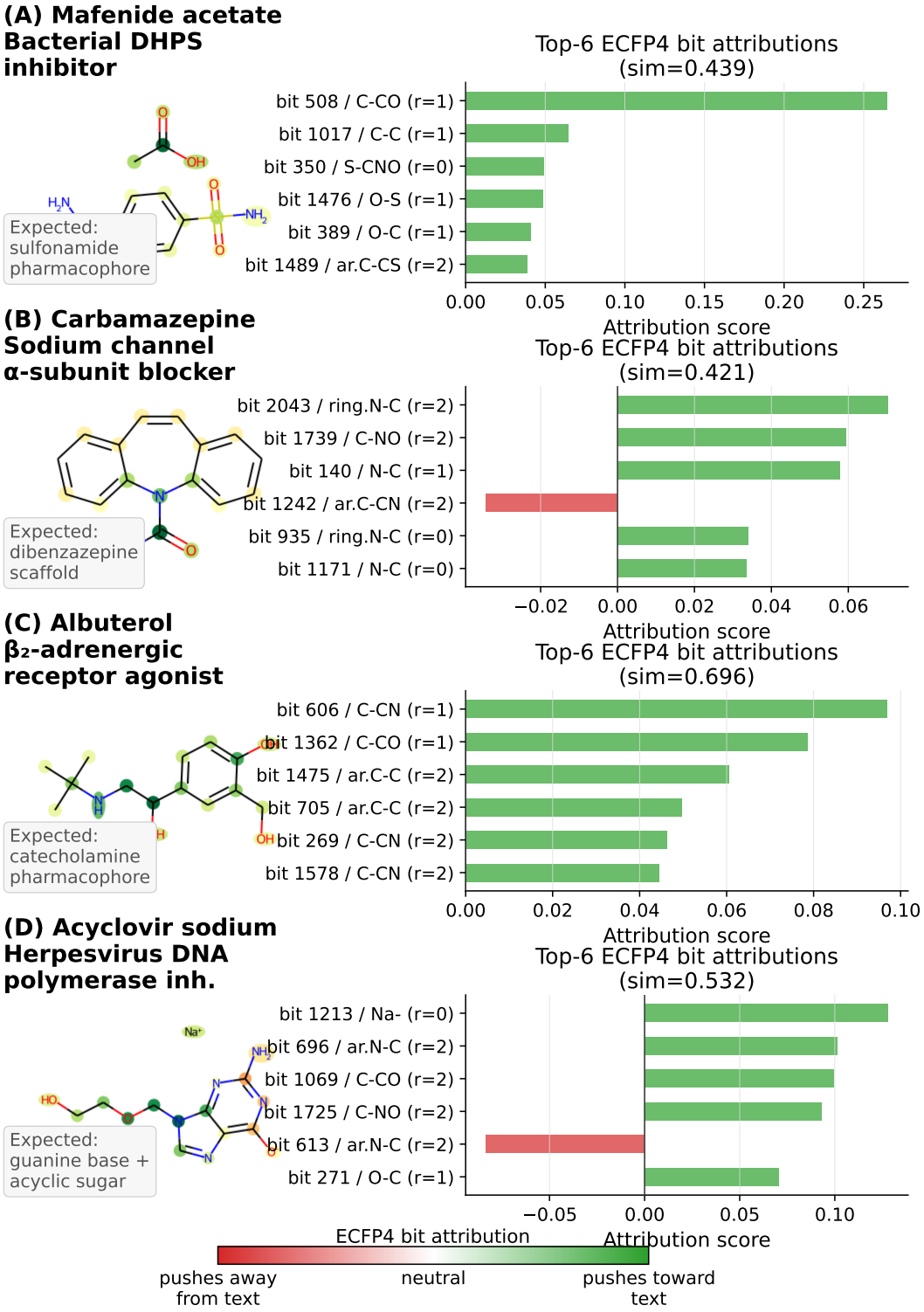

Figure 3: **ECFP4 linear saliency examples for correctly retrieved pairs.** Atoms are colour-coded by attribution value and paired with top-bit bar charts, showing that high-saliency bits often align with mechanism-relevant pharmacophoric regions.

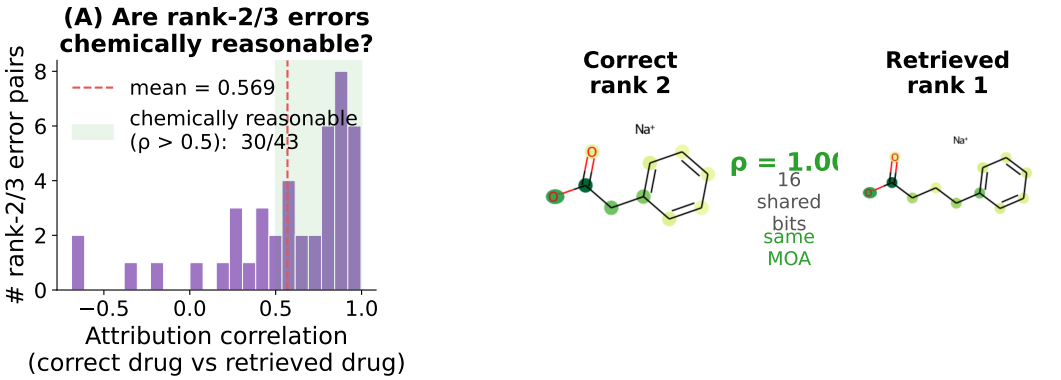

Figure 4: **Auditing wrong-but-close retrievals.** For rank-2/3 errors, attribution correlation separates chemically related near misses from less interpretable failures. Sodium phenylacetate and sodium phenylbutyrate illustrate a high-correlation same-mechanism confusion.

Same-model leakage attribution. We next ask whether the same rich model uses the same substructures when the explicit drug name is removed at inference. For each test pair, let $\mathcal{T}_i^{\text{rich}}$ and $\mathcal{T}_i^{\text{nodrug}}$ be the top-10 saliency-bit sets under the original query and the same query with `Drug: X.` removed. We compute

$$J_i = \frac{|\mathcal{T}_i^{\text{rich}} \cap \mathcal{T}_i^{\text{nodrug}}|}{|\mathcal{T}_i^{\text{rich}} \cup \mathcal{T}_i^{\text{nodrug}}|}. \tag{5}$$

Mean overlap is only 0.356 ± 0.161; 82/435 pairs have $J_i < 0.2$, 36/435 have $J_i > 0.6$, and 13/435 are leakage candidates that are correct under `text_rich`, wrong under the same-model no-name query, and have $J_i < 0.2$. This converts the score-level leakage drop into a structural claim: for a substantial subset of test pairs, removing the name changes which molecular features drive retrieval even though model weights are fixed.

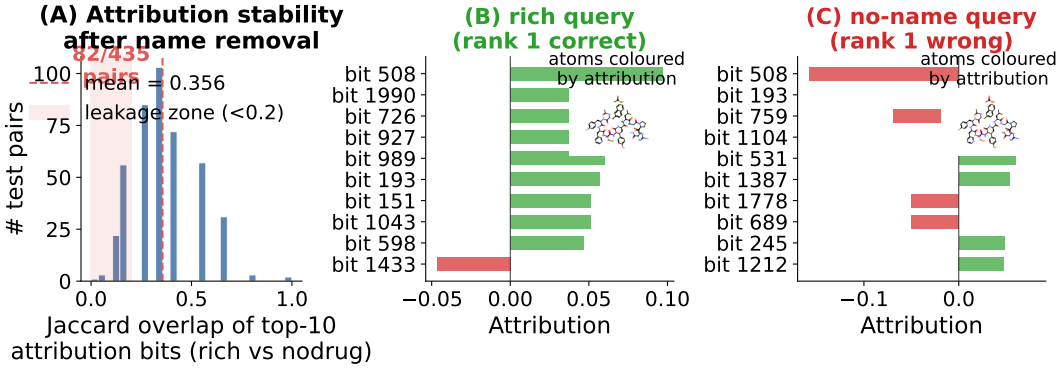

Figure 5: **Same-model drug-name ablation shifts salient substructures.** Removing only the explicit drug-name span from the query changes top-saliency ECFP4 bits for many test pairs; Cetrorelix acetate illustrates a low-overlap leakage candidate.

3.3 Layer-Wise Text Diagnostic

We adapt the logit-lens idea [nostalgebraist, 2020] by applying the trained text projection head W_t to the CLS state at each S-Biomed-RoBERTa layer. The diagnostic exactly replays the main rich model at layer 12: the reconstructed embeddings match the saved outputs within 1.21×10^{-3} maximum absolute error and recover R@1 = 0.225. Layers 0–11 are near chance for both rich and retrained no-drug text (R@1 between 0.002 and 0.011), while layer 12 jumps to R@1 = 0.225 for `text_rich` and 0.113 for `text_nodrug`. This is best viewed as a descriptive sanity check: retrieval signal and the rich-vs-no-name gap both appear in the final contextualized CLS representation, not gradually across intermediate layers. The full layer table and attention visualizations are in Appendix E.

Layer-12 retrieval is heterogeneous across mechanism families rather than uniformly strong (Table 5). The largest gains appear for nuclear-receptor, kinase, antiviral, and antibacterial mechanisms, while

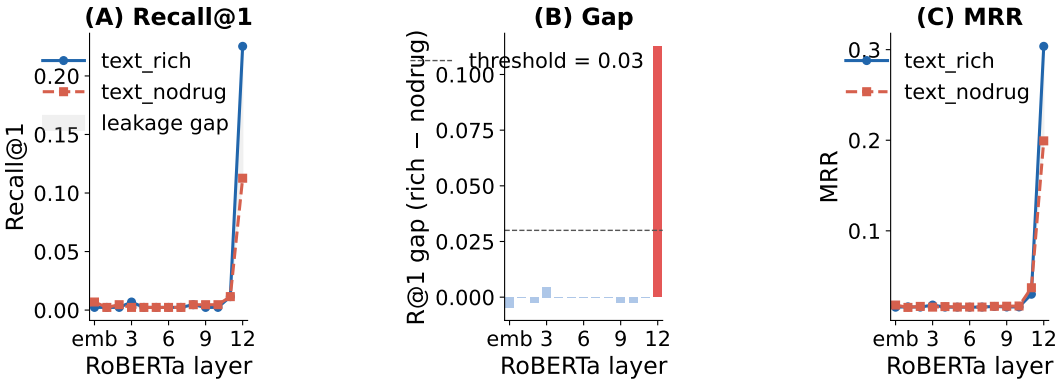

Figure 6: **Contrastive logit lens across RoBERTa layers.** Retrieval is near chance until the final transformer layer, where both rich retrieval signal and the no-name gap emerge.

Table 5: Peak Recall@1 at layer 12 by mechanism family under `text_rich`.

Mechanism family	n	R@1
Nuclear Receptor agonist/antagonist	18	0.611
Kinase inhibitor	11	0.545
Antiviral (DNA polymerase inhibitor)	15	0.533
Antibacterial	25	0.440
Diagnostic agent	15	0.333
GPCR agonist/antagonist	129	0.271
COX Inhibitor	16	0.250
Ion Channel blocker	41	0.195
Other	165	0.152

larger and more heterogeneous families such as GPCR and Other remain harder. We interpret this as evidence that the final-layer signal is shaped by curated mechanism text and family-specific lexical structure, not as a claim that intermediate RoBERTa layers encode chemically useful retrieval features.

4 Discussion and Limitations

MoleculeLens is not a claim that frozen ECFP4 plus a linear bridge solves scaffold-generalized drug–text retrieval. Its narrower contribution is that a lightweight model remains competitive at top-rank scaffold retrieval while making its decisions inspectable at the ECFP4-substructure level. The saliency proxy is validated against exact score-derived quantities, remains faithful on retrieval failures, and exposes same-model drug-name leakage as a shift in salient molecular features.

The benchmark has important limitations. The rich text field includes target and action metadata, and the mechanism-only ablation confirms that these fields contribute retrieval signal; the rich-text result should therefore be interpreted as curated mechanism-metadata retrieval rather than pure free-text mechanism understanding. Implicit lexical cues, such as proprietary-name fragments embedded in mechanism text, are not removed by the no-drug ablation and may still inflate no-drug scores. The dataset covers FDA-approved drugs with curated ChEMBL mechanisms and does not establish generalization to investigational compounds or non-ChEMBL prose. Finally, the baseline set does not include adapted MolT5, BioT5, Uni-Mol, MolCA, or chemistry-LLM retrieval systems; these require nontrivial task adaptation and remain important future comparisons.

Reproducibility and intended use. The benchmark is designed to make these limitations visible rather than hide them. The scaffold split is released as a fixed artifact, the paper-facing tables are regenerated through a single artifact-refresh path, and all main numbers are tied to saved manifests. This matters because retrieval benchmarks are easy to contaminate: changing from scaffold-held-out evaluation to a full-gallery setting, or changing from same-model ablation to retrained no-drug comparison, answers a different scientific question. We therefore report the stronger rich-text result, the stricter mechanism-only condition, and the same-model no-name ablation side by side.

MoleculeLens is intended as an auditing tool for drug–text alignment, not as a standalone decision system for chemistry or medicine. It does not generate new molecules and should not be used for clinical or regulatory decisions without expert review. Its positive use case is narrower: when a retrieval model proposes a structure–mechanism match, MoleculeLens can show which fingerprint environments contributed to that match and whether those environments change under text perturbations. That makes it useful for diagnosing shortcut learning, prioritizing cases for medicinal-chemist inspection, and building stronger future benchmarks where higher-capacity encoders can be evaluated with the same scaffold and leakage controls.

Future interpretability directions. The shared 256-dimensional embedding space also creates a natural target for sparse autoencoder analysis [Templeton et al., 2024, Bricken et al., 2023] to test whether alignment dimensions are monosemantic or polysemantic. A cross-modal crosscoder [Lindsey et al., 2025] trained on paired molecule–text embeddings could further separate features that genuinely align structure and mechanism from features that mostly encode text-only leakage. These extensions are outside the present benchmark, but they follow directly from the linear-bridge design: MoleculeLens supplies a reproducible scaffold split, saved embeddings, and substructure-level saliency targets that stronger encoders and richer interpretability tools can reuse.

Benchmark scope. The benchmark covers FDA-approved drugs with curated ChEMBL mechanism text, deliberately including entries with parseable SMILES and non-empty mechanism descriptions. It does not cover investigational compounds, natural products without approved status, or mechanism texts written in a different domain from the training distribution of S-Biomed-RoBERTa. Same-modality text lookup controls are also not treated as competing baselines because they access gallery text rather than molecular structure; they are better interpreted as text-identifiability audits reported separately from cross-modal retrieval.

5 Conclusion

On a 2,699-pair ChEMBL benchmark with scaffold-disjoint evaluation, MoleculeLens outperforms MolPrompt and KV-PLM on Recall@1 and MRR while clearly exposing the role of lexical and target/action metadata. Its central value is interpretability: the linear molecular bridge yields a validated, efficient ECFP4 saliency lens that can audit correct matches, failed retrievals, and same-model leakage. This makes MoleculeLens a useful foundation for future work on stronger encoders, text-identifiability audits, and feature-level analysis of drug–text alignment.

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A Additional Benchmark and Evaluation Details

The primary retrieval direction is text query to molecule gallery because this is the direction used by the attribution and logit-lens analyses. The diagonal evaluation protocol and parent-structure handling are summarized in the main text.

B Architecture and Implementation Details

The model hyperparameters, runtime, and architecture diagram are summarized in the main text.

C Robustness Details

The corresponding full-loss five-seed check is reported in the main text in Table 3; the robustness summary figure is shown in Figure 2.

D Additional Attribution Analyses

For each test pair, we compute Equation 3, restrict to ECFP4 bits present in the molecule, rank bits by absolute saliency, and decode top-ranked bits with RDKit. Among eight correctly retrieved examples spanning antibacterial, antiviral, nuclear-receptor, opioid, sulfonamide, and diagnostic-agent mechanisms, the highest-saliency bits correspond to pharmacophoric or mechanism-defining regions such as steroid cores, sulfonamide groups, nucleoside analogue motifs, beta-lactam rings, and iodinated benzoate rings.

Table 6: Multi-seed scaffold split variance for MoleculeLens using the simplified proxy objective. Five independent Bemis–Murcko splits are used; only linear projection heads are retrained per seed. Leakage drop is $(R@1_{\text{rich}} - R@1_{\text{nodrug}})/R@1_{\text{rich}}$.

Seed	n_{test}	R@1	R@5	R@10	MRR	Leakage drop
0	403	0.191	0.377	0.459	0.287	29.9%
1	261	0.226	0.479	0.544	0.342	20.3%
2	224	0.188	0.393	0.496	0.296	2.4%
3	221	0.285	0.525	0.634	0.402	27.0%
4	395	0.175	0.352	0.446	0.269	34.8%
Mean	–	**0.213**	**0.425**	**0.516**	**0.319**	**22.9%**
Std	–	0.045	0.070	0.073	0.053	12.6%

Figure 3 shows representative correct retrievals. Aggregate high-saliency bit frequencies are reported in Table 7.

Table 7: Most frequent high-saliency ECFP4 bits across 435 scaffold test pairs. Frequency is the number of test pairs for which the bit appears in the top-10 saliency ranking.

Bit index	Frequency	Mean attribution
1538	85 (19.5%)	0.228
1380	74 (17.0%)	0.004
1873	55 (12.6%)	0.000
807	44 (10.1%)	0.002
715	36 (8.3%)	0.029
350	35 (8.0%)	0.021
1171	33 (7.6%)	0.009
1917	32 (7.4%)	0.001
1039	31 (7.1%)	0.011
80	30 (6.9%)	0.000

The same-model drug-name ablation figure is shown in Figure 5.

E Contrastive Logit-Lens Details

Table 8: Recall@1 and MRR at key RoBERTa layers under both text conditions. Layers 0–11 carry essentially no drug-retrievable signal; meaningful retrieval appears at layer 12.

Layer	text_rich		text_nodrug		Gap
	R@1	MRR	R@1	MRR	R@1
0	0.002	0.016	0.007	0.018	–0.005
3	0.007	0.018	0.002	0.016	0.005
6	0.002	0.016	0.002	0.016	0.000
9	0.002	0.016	0.005	0.017	–0.002
11	0.011	0.030	0.011	0.037	0.000
12	**0.225**	**0.304**	**0.113**	**0.199**	**0.113**

Mechanism-family Recall@1 is reported in the main text in Table 5.

F Broader Impacts

Broader impacts and intended-use considerations are discussed in the main text. In brief, MoleculeLens is an auditing framework for publicly available ChEMBL drug–mechanism retrieval and is not intended for clinical or regulatory deployment without expert oversight.

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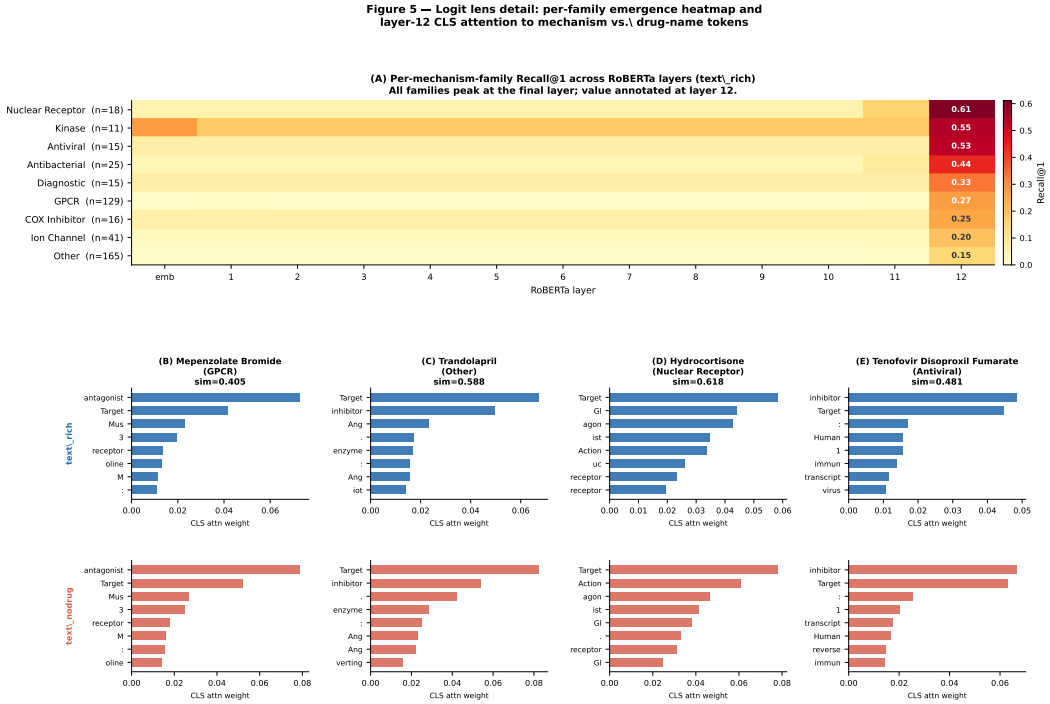

Figure 7: **Logit-lens detail.** Family-level layer-12 retrieval and CLS attention examples under rich and no-drug text conditions.

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LR, weight decay, temperature, margin, epochs), and Section 7.1 states the linear saliency formula and its approximation assumptions. The repo now includes a single refresh command, `bash scripts/run_paper_artifact_refresh.sh`, which rewrites the comparison manifest, canonical Track 1/Track 3 artifacts, paper figures, and a machine-readable `results/paper_artifact_manifest.json` that records the source files and key numbers backing the manuscript.

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